Brownian motion and stochastic integrals, viewed from Ito's isometry and Girsanov's theorem [DRAFT. Thoughts only]

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There are two dual approaches to this that I've observed. One concerns Stochastic processes, the other — distributions. In this article I try to carry out a pedagogical presentation of material studied at uni, i.e. in a way that feels as "natural" as possible rather than forced.

1 Ultra Basics

1.1 Brownian motion

Brownian motion is defined either

- as a stochastic process $(W_t)_{t\geq 0}$ satisfying three properties:
  1. $W_0 = 0$
  2. W_t is normally distributed with variance t; and all increments (of the same length) $\Delta W_{s,t}$ share that same distribution;
     Later I will write ΔW_t to mean $\Delta W_{0,t}$, so just W_t, but written in the increment style to stress that we are interested in increments of a particular length. By stationarity, expected values then are the same for increments not started at zero, too.
  3. For $s < t$, $W_s \perp \Delta W_{s,t}$, i.e. independent increments
- as a limiting distribution on the space of continuous functions $C(\Omega, \mathbb{R}^n)$ of the distributions of random walks on uniform grid $\{t_i \coloneqq i/n\}_{i=0}^{n}$ with jump $1/\sqrt{n}$.

The first view explains how W_t is moving in space, the second — why the variance is $1/t$, i.e. why deviation is $1/\sqrt{t}$.

1.2 Stochastic integral for simple functions

In a Lebesgue-Stieltjes integral style we can integrate simple predictable functions against the brownian motion.

Remember that the Lebesgue-Stieltjes integral against a function (of bounded variation) g,

$$\int_X f \mathrm{d}g$$

works similarly to the standard Lebesgue integral but instead of using a *measure* for the floor increments ($\mu(x-y)$) when infinitesimally multiplying the integrated function, it uses the provided function g's increment, $\Delta g = g(x) - g(y)$, to "measure" it.

A simple process is a process taking finitely many values that we denote $f_1, \ldots, f_k$, i.e. it is expressible as a finite sum

$$f^\omega(t) = \sum_n f_n^\omega \mathbf{1}_{[t_n, t_{n+1}]}(t)$$

So we define the stochastic integral against the brownian motion $\mathrm{d}W_s$ of simple predictable functions as

$$\int_0^t f(s)\mathrm{d}W_s := \sum_n f_n \Delta W_n$$

2 Square of $\triangle$output is expected to be the $\triangle$input

There are a range of facts that I summarize under this section's title. The chaos of the material is vastly reduced by viewing many results as an instance of this pattern.

2.1 Of W_t itself

2.1.1 Discretely

The second view of Brownian motion above also tells us that considering one jump of the random walk, say

$$\text{time: } t_i \to t_{i+1}$$
$$\text{space: } \xi_i \to \xi_{i+1}.$$

With the choice $\Delta t_i = 1/n$, $\Delta \xi_i$ is either $\frac{1}{\sqrt{n}}$ or $-\frac{1}{\sqrt{n}}$. Thus the square $\Delta \xi_i^2$ is most definitely $\frac{1}{n}$. That is,

$$\Delta \xi_i^2 = \Delta t_t$$

— the square of the output's increment equals the input increment.

2.1.2 Continuously

In the limiting distribution $\Delta W_{s,t}$ is definitely does not square to just $(t-s)$, but it is interesting that its mean *does*:

$$\mathbb{E}\left(\Delta W_t^2\right) = t.$$

We see this from the property charaterising the distribution of the (non-squared) increment ΔW_t:

$$\begin{aligned}\mathbb{E}\left(\Delta W_t^2\right) &\equiv \int_\Omega \Delta W_t^2 \mathrm{d}\mathbb{P} = \int_\mathbb{R} y^2 \mathrm{d}\mathbb{P}_Z \\ &= \int_\mathbb{R} y^2 \Phi(y) \mathrm{d}y \overset{!}{=} \sigma^2 \int_\mathbb{R} \Phi(y)\mathrm{d}y = t\end{aligned}$$

where in $\overset{!}{=}$ we used (σ^2 denoting the variance of the normal variable, i.e. here just t; also omitting integral signs)

$$\begin{aligned}y^2 e^{-y^2/2\sigma^2}\mathrm{d}y &= -2\sigma^2 y e^{-y^2/2\sigma^2}\mathrm{d}y^2 / -2\sigma^2 \\ &= -y\mathrm{d}e^{-y^2/2\sigma^2} \\ &= -ye^{-y^2/2\sigma^2} + e^{-y^2/2\sigma^2}\mathrm{d}y.\end{aligned}$$

Thoughts

Note:::: I can't quite explain yet why in the discrete case randomness goes away ($\Delta\xi_i^2$ is not random), while in the continuous – not (ΔW_t^2 is still random).

2.2 Of the Ito integral for simple functions

Above we computed the expected square of the raw increments ΔW_t. In a martingale-transformation fashion we can consider the "transformed" increments $f\Delta W$ and their integral's expected square:

$$\begin{aligned}\mathbb{E}\left[\left(\int_0^t f(s)\mathrm{d}W_s\right)^2\right] &\equiv \mathbb{E}\left[\left(\sum\nolimits_n f_n \Delta W_n\right)^2\right] \\ &\overset{\text{lin}}{=} \mathbb{E}\left[\sum_n f_n^2 \Delta W_n^2\right] + \mathbb{E}\left[\sum_{n\neq m} f_n f_m \Delta W_n \Delta W_m\right] \\ &\overset{\perp}{=} \sum_n \mathbb{E}\left[f_n^2\right] \underbrace{\mathbb{E}\left[\Delta W_n^2\right]}_{\Delta t} \cancel{+2\sum_{n<m} \mathbb{E}\left[f_n f_m \Delta W_n\right] \underbrace{\mathbb{E}\left[\Delta W_m\right]}_{0}} \\ &= \mathbb{E}\left[\sum_n f_n^2 \Delta t\right] \\ &\equiv \mathop{\mathbb{E}}_\omega\left[\int_0^t f_\omega^2(s)\mathrm{d}s\right]\end{aligned}$$

which I roughly interpret as

> The expected **square** of the modified (by f_t) output (of ΔW_t) equals the **linear** integral of the output's modification's square.

In a way, the ΔW_t^2 collapses inside the integral to Δt, while the f^2 multiple remains as a squre.

This peculiar fact carries the name "Ito's Isometry" (for the very special case of simple functions).